Chapter 7. Analysis of Discrete Data

Proportions and contingency tables

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7.1 Estimation and testing of a population proportion

Motivation. Many quantities of interest — defect rates, unemployment rates, approval ratings — are proportions: the fraction of a population having some attribute. The count of such units in a random sample is binomial, and for large samples the normal approximation gives confidence intervals and tests.

Let p be the proportion of an (infinite) population having an attribute, and let X be the number of units with that attribute in a random sample of size n. Then $X \sim B(n, p)$, and the natural estimator of p is the **sample proportion**

$$\hat{p} = \frac{X}{n}, \qquad \mathrm{Var}(\hat{p}) = \frac{\mathrm{Var}(X)}{n^2} = \frac{p(1-p)}{n}.$$

The standard error $\mathrm{SE}(\hat{p}) = \sqrt{p(1-p)/n}$ depends on the unknown p, so it is estimated by plugging in $\hat{p}$:

$$\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

By the normal approximation to the binomial, $X \approx N(np, np(1-p))$ for large n, and

$$\frac{X - np}{\sqrt{n\hat{p}(1-\hat{p})}} = \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \;\approx\; N(0,1), \qquad \text{so} \qquad P\Big(-z_{\alpha/2} \le \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \le z_{\alpha/2}\Big) \approx 1 - \alpha.$$

Definition (Estimating p (large sample))

(i) Estimator and standard error: $\hat{p} = X/n$, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$.
(ii) $100(1-\alpha)\%$ margin of error: $z_{\alpha/2}\,\widehat{\mathrm{SE}}(\hat{p})$.
(iii) $100(1-\alpha)\%$ confidence interval:

$$\Big(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\,,\;\; \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\Big).$$

Rule of thumb

The normal approximation is usually considered adequate when $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$.

Example 7-1 (Homecoming rate in Seoul)

To estimate the proportion p of Seoul households that will travel home for Chuseok, 500 households were sampled by simple random sampling; 79 said they would go. Find a 95% CI for p.
Solution. The population is huge, so treat the sample as a random sample from an infinite population. $\hat{p} = 79/500 = 0.158$, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{0.158(0.842)/500} = 0.016$. Since $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$, the 95% CI is

$$(0.158 - 1.96 \times 0.016,\; 0.158 + 1.96 \times 0.016) = (0.126,\; 0.190).$$

7.1.1 Choosing the sample size

To keep the $100(1-\alpha)\%$ margin of error at most d we need $z_{\alpha/2}\sqrt{p(1-p)/n} \le d$. Since p is unknown and $p(1-p) \le 1/4$ for all p, it suffices that $z_{\alpha/2}\sqrt{1/4}/\sqrt{n} \le d$.

Definition (Sample size for estimating p)

To make the margin of error at most d (equivalently, the CI length at most $2d$):
(i) with a prior estimate p^* of p: $n \geq p^*(1-p^*)(z_{\alpha/2}/d)^2$;
(ii) with no prior information: $n \geq \frac{1}{4}(z_{\alpha/2}/d)^2$.

Why (ii) suffices. $p(1-p) = \frac{1}{4} - (p-\frac{1}{2})^2 \leq \frac{1}{4}$, so if $n \geq \frac{1}{4}(z_{\alpha/2}/d)^2$ then $z_{\alpha/2}\sqrt{p(1-p)/n} \leq z_{\alpha/2}\sqrt{1/(4n)} \leq d$ whatever the true p. ■

Example 7-2 (Poll for a district election)

Two candidates A and B run for district mayor. With no prior information, how large a sample is needed so that the 95% margin of error for A's support rate is within 3%?
Solution. $n \geq \frac{1}{4}(1.96/0.03)^2 = 1067.1$, so sample $n = 1068$ people.

7.1.2 Tests for p

Consider $H_0 : p = p_0$ vs. $H_1 : p > p_0$. Large observed counts x are evidence against H_0, so the P-value is $P = P(X \geq x)$ with $X \sim B(n, p_0)$, computed exactly from the binomial; reject at level α iff $P \leq \alpha$.

Definition (Small-sample test for p)

Test statistic: the count X, observed value x; $X \sim B(n, p_0)$ under $H_0 : p = p_0$.

H_1	P-value	reject at level α
(a) $p > p_0$	$P(X \geq x)$	$P \leq \alpha$
(b) $p < p_0$	$P(X \leq x)$	$P \leq \alpha$
(c) $p \neq p_0$	$P(X \leq y(x)) + P(X \geq x)$ if $x \geq np_0$	$P \leq \alpha$
	$P(X \leq x) + P(X \geq y(x))$ if $x < np_0$	

In (c), $y(x)$ is the integer on the opposite side of np_0 from x that is the largest ($x \geq np_0$) or smallest ($x < np_0$) nonnegative integer k with $P(X = k) \leq P(X = x)$.

Since the exact P-value is practical only for small n, this is called the **small-sample test** for a proportion.

Example 7-3 (A cheaper cutting tool)

With the current tool the defect rate is 5%. A cheaper tool may raise it, so 20 items were produced with it; 3 were defective. Test at level 5%.
Solution. $H_0 : p = 0.05$ vs. $H_1 : p > 0.05$, $n = 20$, $x = 3$. $P = P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.925 = 0.075$ for $X \sim B(20, 0.05)$. As $0.075 > 0.05$, H_0 is not rejected: this trial does not show that the cheaper tool raises the defect rate.

For large n, under H_0,

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \approx N(0,1),$$

usable when $np_0 \geq 5$ and $n(1-p_0) \geq 5$. Note that the null value p_0 (not $\hat{p}$) is used in the standard error.

Definition (Large-sample test for p ($np_0 \geq 5$, $n(1-p_0) \geq 5$))

$Z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$ with observed value z.

H_1	P-value	rejection region
(a) $p > p_0$	$P(Z \geq z)$	$z \geq z_\alpha$
(b) $p < p_0$	$P(Z \leq z)$	$z \leq -z_\alpha$
(c) $p \neq p_0$	$P(\|Z\| \geq \|z\|)$	$\|z\| \geq z_{\alpha/2}$

Example 7-4 (Has the homecoming rate dropped?)

In the past the Chuseok homecoming rate was 20%. Using the data of Example 7-1, test at level 1% whether the rate has decreased.
Solution. $H_0 : p = 0.2$ vs. $H_1 : p < 0.2$; $n = 500$, $X = 79$, $np_0 = 100 \geq 5$, $n(1-p_0) = 400 \geq 5$.

$$z = \frac{79/500 - 0.2}{\sqrt{0.2 \times 0.8/500}} = -2.348 \;\leq\; -z_{0.01} = -2.327,$$

so H_0 is rejected at level 1%: the homecoming rate is significantly lower than before.

7.2 Comparing two population proportions

Motivation. Comparing cure rates of two treatments or defect rates of two processes requires inference on $p_1 - p_2$ from two independent samples; the same normal approximation, applied to the difference, does the job.

Independent random samples of sizes n_1, n_2 are drawn from two populations; $X_1 \sim B(n_1, p_1)$ and $X_2 \sim B(n_2, p_2)$ are independent. The estimator of $p_1 - p_2$ is

$$\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2}, \qquad \mathrm{Var}(\hat{p}_1 - \hat{p}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2},$$

by independence. For large n_1, n_2, $\hat{p}_i \approx N(p_i, p_i(1-p_i)/n_i)$ independently, so $\hat{p}_1 - \hat{p}_2$ is approximately normal and

$$\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}} \;\approx\; N(0,1).$$

Definition (Estimating $p_1 - p_2$ (large n_1, n_2))

(i) $\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2$, $\widehat{\mathrm{SE}} = \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$.
(ii) $100(1-\alpha)\%$ CI for $p_1 - p_2$: $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\,\widehat{\mathrm{SE}}(\widehat{p_1 - p_2})$.

Example 7-5 (1954 Salk polio vaccine trial)

In a double-blind trial, a control group received placebo and a treatment group the vaccine. Let p_1, p_2 be the polio rates in the control and treatment groups.

	sample size	children with polio
Control	201,229	115
Treatment	200,745	33

Estimate $p_1 - p_2$ and give a 95% CI.

Solution. $\hat p_1 = 115/201229 = 5.71\times10^{-4}$, $\hat p_2 = 33/200745 = 1.64\times10^{-4}$, so $\hat p_1-\hat p_2 = 4.07\times10^{-4}$ with $\widehat{\mathrm{SE}} = 0.60\times10^{-4}$. The 95% CI is $4.07\times10^{-4}\pm1.96\times0.60\times10^{-4}$, i.e.

$$2.89\times10^{-4} < p_1 - p_2 < 5.25\times10^{-4}.$$

7.2.1 Testing $H_0: p_1 = p_2$

Under H_0 write $p = p_1 = p_2$; then $\mathrm{Var}(\hat p_1-\hat p_2) = p(1-p)(1/n_1+1/n_2)$. Pooling the two samples, X_1+X_2 of the n_1+n_2 units have the attribute, so p is estimated by the **pooled sample proportion**

$$\hat p = \frac{X_1+X_2}{n_1+n_2}, \qquad \widehat{\mathrm{Var}}(\hat p_1-\hat p_2) = \hat p(1-\hat p)\Big(\frac{1}{n_1}+\frac{1}{n_2}\Big).$$

Definition (Large-sample test of $H_0: p_1 = p_2$)

$$Z = \frac{\hat p_1-\hat p_2}{\sqrt{\hat p(1-\hat p)(n_1^{-1}+n_2^{-1})}}, \qquad \hat p = \frac{X_1+X_2}{n_1+n_2}.$$

With observed value z and $Z\approx N(0,1)$ under H_0:
(a) $H_1: p_1 > p_2$: $P = P(Z\ge z)$, reject if $z\ge z_\alpha$;
(b) $H_1: p_1 < p_2$: $P = P(Z\le z)$, reject if $z\le -z_\alpha$;
(c) $H_1: p_1 \ne p_2$: $P = P(|Z|\ge|z|)$, reject if $|z|\ge z_{\alpha/2}$.

Example 7-6 (Is the vaccine effective?)

With the data of Example 7-5, test whether the vaccine is effective.
Solution. $H_0: p_1 = p_2$ vs. $H_1: p_1 > p_2$ (p_1 placebo, p_2 vaccine). $\hat p = (115+33)/(201229+200745) = 3.68\times10^{-4}$,

$$z = \frac{5.71\times10^{-4}-1.64\times10^{-4}}{\sqrt{\hat p(1-\hat p)(n_1^{-1}+n_2^{-1})}} = 6.727, \qquad P = P(Z\ge6.727) = 8.66\times10^{-12}.$$

H_0 is rejected decisively: the 1954 trial strongly indicates that the vaccine prevents polio.

Example 7-7 (Parole success of murderers)

Random samples of 42 impulsive and 40 premeditated murderers released on parole were followed; success means no reoffence. Test at level 5% whether the success rates differ.

	success	failure	size
Impulsive	13	29	$n_1 = 42$
Premeditated	22	18	$n_2 = 40$
Total	35	47	$n = 82$

Solution. $H_0: p_1 = p_2$ vs. $H_1: p_1 \ne p_2$. $\hat p_1 = 13/42 = 0.310$, $\hat p_2 = 22/40 = 0.550$, $\hat p = 35/82 = 0.427$, and

$$z = \frac{\hat p_1-\hat p_2}{\sqrt{\hat p(1-\hat p)(n_1^{-1}+n_2^{-1})}} = -2.20, \qquad |z|\ge z_{0.025} = 1.96,$$

so H_0 is rejected: the parole success rates differ significantly.

7.3 Comparing several populations with categorical data

Motivation. When there are more than two populations, or more than two categories, the z-test no longer applies. Comparing observed counts with the counts expected under H_0 leads to the chi-square statistic, which handles any number of populations and categories.

The idea: reject H_0 (all populations have the same category proportions) when the **observed frequencies** differ much from the **estimated expected frequencies** under H_0, where

$$\text{estimated expected frequency} = (\text{sample size}) \times (\text{estimated probability under } H_0).$$

Illustration 7-1: expected counts for Example 7-7

Under $H_0 : p_1 = p_2$ the common success rate is estimated by $\hat{p} = 0.427$, so

$$\hat{E}_{11} = n_1\hat{p} = 42 \times 0.427 = 17.93, \qquad \hat{E}_{12} = n_1(1-\hat{p}) = 24.07,$$
$$\hat{E}_{21} = n_2\hat{p} = 40 \times 0.427 = 17.08, \qquad \hat{E}_{22} = n_2(1-\hat{p}) = 22.92.$$

Observed (expected): impulsive 13 (17.93), 29 (24.07); premeditated 22 (17.08), 18 (22.92). The discrepancy statistic is

$$\sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}} = \frac{(13-17.93)^2}{17.93} + \cdots + \frac{(18-22.92)^2}{22.92} = 4.84 = (-2.20)^2 = z^2.$$

The identity $\sum\sum(O_{ij}-\hat{E}_{ij})^2/\hat{E}_{ij} = z^2$ holds in general for 2×2 tables; since $Z \approx N(0,1)$ under H_0, the statistic is approximately $\chi^2(1)$. It is called the **chi-square statistic**,

$$\chi^2 = \sum_i\sum_j \frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}}.$$

In Example 7-7 the 5% critical region is $\chi^2 \geq \chi^2_{0.05}(1) = 3.84$, equivalently $z^2 \geq z^2_{0.025} = 1.96^2 = 3.84$.

Fact ($\chi^2 = z^2$ for a 2×2 table)

For two samples with pooled proportion $\hat{p}$, the chi-square statistic of the 2×2 table equals the square of the two-sample z-statistic.

Proof. [Supplement] Write $q = 1 - \hat{p}$ and $D = X_1 - n_1\hat{p}$. Because $X_1 + X_2 = (n_1 + n_2)\hat{p}$, we have $X_2 - n_2\hat{p} = -D$, and for the failures $(n_i - X_i) - n_iq = -(X_i - n_i\hat{p})$. Hence every $|O_{ij} - \hat{E}_{ij}| = |D|$, and

$$\chi^2 = D^2\Big(\frac{1}{n_1\hat{p}} + \frac{1}{n_1q} + \frac{1}{n_2\hat{p}} + \frac{1}{n_2q}\Big) = \frac{D^2}{\hat{p}q}\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big).$$

On the other hand $\hat{p}_1 - \hat{p}_2 = D/n_1 + D/n_2 = D(n_1^{-1} + n_2^{-1})$, so

$$z^2 = \frac{D^2(n_1^{-1}+n_2^{-1})^2}{\hat{p}q(n_1^{-1}+n_2^{-1})} = \frac{D^2}{\hat{p}q}\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big) = \chi^2.$$

■

7.3.1 Several binomial populations

Observing r binomial populations gives an $r \times 2$ **contingency table** with counts O_{i1} (success), O_{i2} (failure), row totals n_i, column totals $O_{\cdot 1}, O_{\cdot 2}$ and grand total n. We test

$$H_0 : p_1 = p_2 = \cdots = p_r \qquad \text{vs.} \qquad H_1 : \text{not all } p_i \text{ are equal.}$$

Under H_0 the common rate is estimated by the pooled proportion $\hat{p} = (O_{11} + \cdots + O_{r1})/(n_1 + \cdots + n_r) = O_{\cdot 1}/n$, and

$$\hat{E}_{i1} = n_i \hat{p} = n_i \frac{O_{\cdot 1}}{n}, \qquad \hat{E}_{i2} = n_i(1-\hat{p}) = n_i \frac{O_{\cdot 2}}{n} \qquad (i = 1, \ldots, r).$$

For large samples, $\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{2} (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij} \approx \chi^2(r-1)$ under H_0 (this is a known result; it rests on the multivariate normal approximation).

Definition (Comparing r binomial populations ($\hat{E}_{ij} \geq 5$))

For $H_0 : p_1 = \cdots = p_r$ vs. H_1: at least one differs, use

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{2} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}, \qquad \hat{E}_{ij} = n_i \frac{O_{\cdot j}}{n}.$$

With observed value χ_0^2: $P = P(\chi^2 \geq \chi_0^2)$ for $\chi^2 \sim \chi^2(r-1)$; reject at level α iff $\chi_0^2 \geq \chi_\alpha^2(r-1)$.

Example 7-8 (Siting an LNG storage base)

Residents of three candidate regions were sampled (400, 350, 350 people) and asked whether they favour an LNG storage base. Do the approval rates differ?

	for	against	size
Region 1	198	202	400
Region 2	140	210	350
Region 3	133	217	350
Total	471	629	1100

Solution. $H_0 : p_1 = p_2 = p_3$. Expected counts: $\hat{E}_{11} = 400 \cdot 471/1100 = 171.27$, $\hat{E}_{12} = 228.73$, $\hat{E}_{21} = \hat{E}_{31} = 149.86$, $\hat{E}_{22} = \hat{E}_{32} = 200.14$. Then

$$\chi_0^2 = \frac{(198 - 171.27)^2}{171.27} + \cdots + \frac{(217 - 200.14)^2}{200.14} = 11.75,$$

and $P = P(\chi^2(2) \geq 11.75) = 0.003$ (from the table, $0.0025 < P < 0.005$). The approval rates differ significantly across the regions.

7.3.2 Several multinomial populations

If each population is classified into c **categories**, we have **multinomial populations**; observing r of them gives an $r \times c$ table of **categorical data**. Let $p_{i1}, \ldots, p_{ic}$ ($\sum_j p_{ij} = 1$) be the category proportions in population i. The test of

$$H_0 : p_{1j} = p_{2j} = \cdots = p_{rj} \ (j = 1, \ldots, c) \qquad \text{vs.} \qquad H_1 : \text{not } H_0$$

is the **test of homogeneity**. Under H_0 the common proportion p_j of category j is estimated by $\hat{p}_j = O_{\cdot j}/n$, so $\hat{E}_{ij} = n_i \hat{p}_j = n_i O_{\cdot j}/n$, and for large samples

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} \approx \chi^2((r-1)(c-1)).$$

Counting degrees of freedom

Each of the r rows has $c-1$ free proportions, giving $r(c-1)$ parameters in general; under H_0 only the $c-1$ common p_j remain. The difference $r(c-1)-(c-1)=(r-1)(c-1)$ is the degrees of freedom. For $c=2$ this is $r-1$, as above.

Definition (Test of homogeneity ($\hat{E}_{ij} \geq 5$))

$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}(O_{ij}-\hat{E}_{ij})^2/\hat{E}_{ij}$ with $\hat{E}_{ij} = n_i O_{\cdot j}/n$. With observed value χ_0^2: $P = P(\chi^2 \geq \chi_0^2)$, $\chi^2 \sim \chi^2((r-1)(c-1))$; reject iff $\chi_0^2 \geq \chi_\alpha^2((r-1)(c-1))$.

Example 7-9 (Odour from an industrial complex)

In three regions next to an industrial complex, 97, 95 and 99 people were asked how often they notice a bad odour: 1 = daily, 2 = at least weekly, 3 = at least monthly, 4 = less than monthly, 5 = never. Test at level 1% whether the frequency differs by region.

	1	2	3	4	5	size
Region 1	20	28	23	14	12	97
Region 2	14	34	21	14	12	95
Region 3	4	12	10	20	53	99
Total	38	74	54	48	77	291

Solution. $\hat{p}_j = 38/291, 74/291, 54/291, 48/291, 77/291$; $\hat{E}_{1j} = 97\hat{p}_j$, $\hat{E}_{2j} = 95\hat{p}_j$, $\hat{E}_{3j} = 99\hat{p}_j$:

	1	2	3	4	5
Region 1	12.67	24.67	18.00	16.00	25.67
Region 2	12.41	24.16	17.63	15.67	25.14
Region 3	12.93	25.18	18.37	16.33	26.20

All expected counts exceed 5. $\chi_0^2 = (20-12.67)^2/12.67 + \cdots + (53-26.20)^2/26.20 = 71.36 \geq \chi_{0.01}^2(8) = 20.09$, so H_0 is rejected: odour perception differs by region.

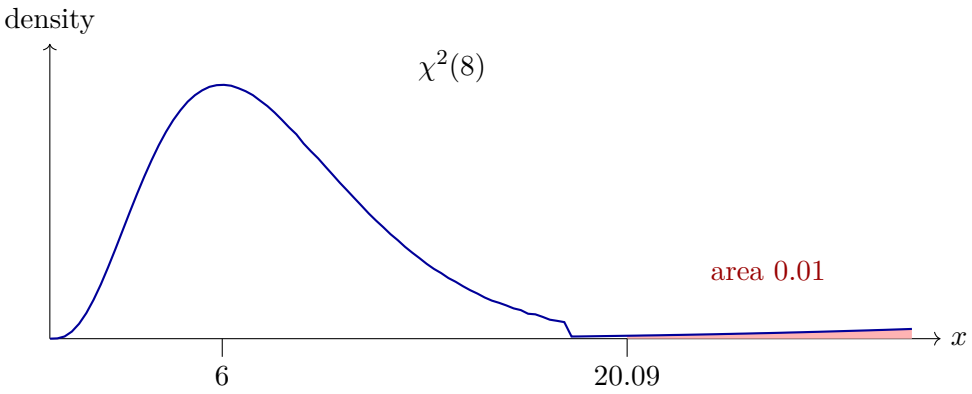


Figure 1: The $\chi^2(8)$ density with the 1% rejection region of Example 7-9; the observed $\chi_0^2 = 71.36$ lies far to the right.

7.4 Test of independence with categorical data

Motivation. Sometimes a single sample is cross-classified by two attributes, and the question is whether the attributes are related. The same chi-square statistic answers it, with expected counts computed from the product of the margins.

Each unit of one population is classified by attribute A (r categories $A_1, \dots, A_r$) and attribute B (c categories $B_1, \dots, B_c$). Let p_{ij} be the cell probabilities, $p_{i\cdot} = \sum_j p_{ij}$ and $p_{\cdot j} = \sum_i p_{ij}$ the marginal probabilities; a random sample of size n gives counts O_{ij} with margins $O_{i\cdot}, O_{\cdot j}$. Independence of A and B means

$$H_0 : p_{ij} = p_{i\cdot}\, p_{\cdot j} \ \text{ for all } i, j, \qquad H_1 : p_{ij} \neq p_{i\cdot}\, p_{\cdot j} \ \text{ for at least one } (i, j).$$

Under H_0 we estimate each margin separately and multiply:

$$\hat{p}_{ij} = \hat{p}_{i\cdot}\, \hat{p}_{\cdot j} = \frac{O_{i\cdot}}{n} \cdot \frac{O_{\cdot j}}{n}, \qquad \hat{E}_{ij} = n\hat{p}_{ij} = \frac{O_{i\cdot}\, O_{\cdot j}}{n}.$$

For large n, $\chi^2 = \sum_i \sum_j (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij} \approx \chi^2((r-1)(c-1))$ under H_0: in general there are $rc - 1$ free cell probabilities, under H_0 only $(r-1) + (c-1)$, and $rc - 1 - (r-1) - (c-1) = (r-1)(c-1)$.

Definition (Test of independence ($\hat{E}_{ij} \geq 5$))

For $H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$ $(i = 1, \dots, r,\ j = 1, \dots, c)$ vs. H_1 : not H_0,

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}, \qquad \hat{E}_{ij} = \frac{O_{i\cdot} O_{\cdot j}}{n}.$$

$P = P(\chi^2 \geq \chi_0^2)$ with $\chi^2 \sim \chi^2((r-1)(c-1))$; reject iff $\chi_0^2 \geq \chi_\alpha^2((r-1)(c-1))$.

Example 7-10 (Car size and commuting distance)

250 solo car commuters in a metropolitan area were classified by car size and commuting distance. Test at level 5% whether they are related.

	<15 km	15–30 km	≥ 30 km	total
Light	6	27	19	52
Small	8	36	17	61
Mid-size	21	45	33	99
Large	14	18	6	38
Total	49	126	75	250

Solution. $\hat{E}_{ij} = O_{i\cdot} O_{\cdot j}/250$: row 1: 10.19, 26.21, 15.60; row 2: 11.96, 30.74, 18.30; row 3: 19.40, 49.90, 29.70; row 4: 7.45, 19.15, 11.40 — all ≥ 5. Then

$$\chi_0^2 = \frac{(6 - 10.19)^2}{10.19} + \cdots + \frac{(6 - 11.40)^2}{11.40} = 14.16 \ \geq \ \chi_{0.05}^2(6) = 12.59,$$

so H_0 is rejected: car size and commuting distance are related.

Homogeneity vs. independence

The test of independence and the test of homogeneity (§7.3) use the same statistic and the same procedure, but the sampling differs: for homogeneity the row totals n_i are fixed in advance (a separate sample from each population); for independence only n is fixed and one sample is cross-classified. Choose the sampling plan that fits the question and then the matching inference.